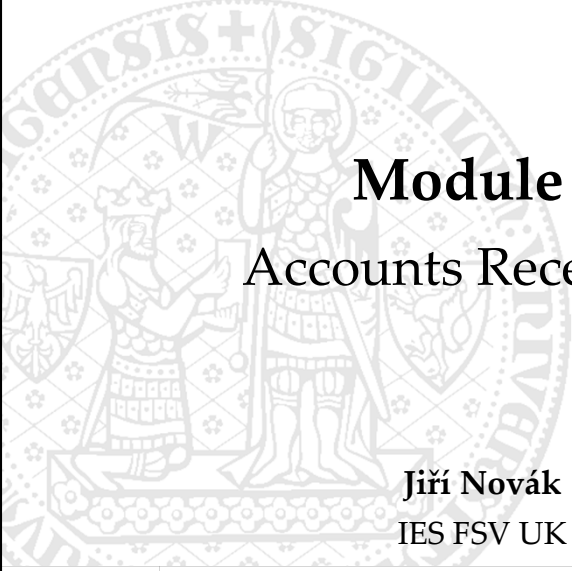




Module 4

Accounts Receivable

Jiří Novák
IES FSV UK



Recap

▪ Inventories

- items held for sale (merchandise) or used in manufacturing (raw materials, work-in-progress, finished goods)
- *specific identification method* – impractical, used rather infrequently (e.g. old cars, antiques)
- *cost flow assumption* – determines how the costs of goods available for sale is distributed between the ending inventory and COGS
 - FIFO (first-in-first-out),
 - LIFO (last-in-first-out),
 - AC (average cost)



Recap

- **Precise Measurement**

- assume purchase prices of inventory tend to rise over time (*inflation*)
 - “new” ~ “LI” – cost of recent inventory units are closest to current prices
 - “old” ~ “FI” – cost of inventory bought long time ago *understates* current cost when
- accounting measurement *more precise* for “LI”
 - LIFO – gross profit (IS),
 - FIFO = LILO – inventory (BS)

- 3 -



Recap

- **Overstating**

- if inventory purchase prices tend to rise, using any *historical* prices:
 - *understates* inventory (BS)
 - *understates* COGS (IS) ~ *overstates* profit (IS)
- *overstating* inventory (BS), COGS (IS) or *understating* profit (IS) would be hypothetically possible only using *future* prices

- 4 -



Recap



■ Measurement Bias

■ FIFO

- old prices in *COGS* that are *understated*
- understated *COGS* ~ *overstated* gross margin

■ LIFO

- old prices in *inventory* that are *understated*
- *LIFO reserve* ($LFR = Iv_F - Iv_M$) artificially low level of inventory cost on balance sheet

	gross profit (IS)	inventory (BS)
FIFO	overstated	OK
LIFO	OK	understated

- 5 -



Recap

■ Why LIFO?

- provides accurate measurement of *COGS* in IS, which is arguably the more important financial statement relative to BS

■ Why FIFO?

- BS distortion due to LIFO *accumulates* over time
- typically reflects *physical flow* of inventory (e.g. fresh milk), but for raw materials (e.g. coal)
- *temporary manipulation* – persistent bias in *COGS* may be less bad than one-off manipulation through liquidation of LIFO reserve

- 6 -



Recap

- **Limit Impact?**
 - How to limit confusion from temporary income manipulation by liquidation of LIFO reserve?
- **Compute Effect?**
 - How to compute magnitude of income improvement by liquidation of LIFO reserve?
- **Trust Investors?**
 - Decision on LIFO depends on the trust in investors' sophistication and meticulousness.

- 7 -

Accounts Receivable



Accounts Receivable



▪ Credit Sales

- most sales are not done for cash but “on credit” (“on account”), i.e. the customer pays later
- credit terms: “2/10, net 30” ... customer must pay in 30 days, gets 2% discount if pays in 10 days

Balance Sheet		Income Statement	
Assets	Liabilities, Equity	Expenses	Revenues
accounts receivable			sales
① 800 ② 700			① 800
cash			
② 700			

- 9 -



Accounts Receivable



▪ Net Realizable Value

- accounts receivable (AR) represent the amount of cash the company expects to collect for its sales, i.e. they must be recorded net of the estimate of sales that will be adjusted for:
 - *bad debt* – customers may sometimes fail to pay for the goods and services they purchased (low priority, no collateral)
 - *sales returns and allowances* – customers may sometimes cancel their purchase or return the goods due to the product guarantee provided

- 10 -



Bad Debts



- 11 -



Bad Debts

- **Direct Write-Off Method**
 - when some receivable is identified as uncollectible *AR* are reduced (*Cr*) and the corresponding bad debt expense (*Dr*) is recognized
- **Allowance Method**
 - allowance for uncollectible accounts (*AUA*) is set up and a related expense recorded already at revenue recognition
 - when receivables are identified as uncollectible *AR* are reduced (*Cr*) against *AUA* (*Dr*)

- 12 -



Bad Debts

▪ Direct Write-Off

- ❶ receivables are recorded at the full amount
- ❷ if some debts are identified as uncollectible, accounts receivable (*As*) are written off against bad debt expense (*Ex*)

	Assets		Liabilities, Equity	Expenses		Revenues	
	Accounts Receivable			Bad Debt Expense		Sales	
❶	800					800	
❷		60		60			
❷ ❶	740						

- 13 -



Bad Debts



▪ Allowance Method

- ❶ allowance for uncollectible accounts (*AUA*) is regularly created at sale
- ❷ as individual *AR* are identified as uncollectible they are eliminated against *AUA*

	Assets		Liabilities, Equity	Expenses		Revenues	
	Accounts Receivable					Sales	
❶	800					800	
❷		60					
	AUA			Bad Debt Expense			
❶		80		80			
❷	60						
❷ ❶	720						

- 14 -



Accounts Receivable



“You handled the audit so well, we’re promoting you from Accounts Receivable to Accounts Deceivable.”

- 15 -



Bad Debts

- **Questions**
 - What is the key *distinction*?
 - What are *pros & cons* of the two methods?
 - Which method is more:
 - *reliable*
 - *relevant*
 - How do the two methods address managerial incentive (dating game vs. begging game)?
 - Which method should regulators require in:
 - *tax* accounting
 - *financial* accounting

- 16 -



Definition

Reliability

“Accounting reliability refers to whether financial information can be verified and used consistently by investors and creditors with the same results. Basically, reliability refers to the trustworthiness of the financial statements.”

MyAccountingCourse.com

- 17 -



Definition

Relevance

“Relevance in accounting means the information we get from the accounting system will help the end-users to make important decisions [...] the information should be useful, understandable, timely, and pertinent for end-users to make important decisions.”

WallStreetMojo.com

- 18 -



Bad Debts

Comparison

- *relevance*
 - allowance method ensures proper *matching* by booking expenses immediately with revenues
- *reliability*
 - direct write-off does not use subjective *estimates* about receivables not to be collected

	direct write-off	allowance
<i>relevance</i>	✗	✓
<i>reliability</i>	✓	✗

- 19 -



Bad Debts

Financial Accounting

- *allowance* method required whenever bad debts can be reasonably estimated:
 - “*dating game*” – using *AUA* prevents managers from overstating net sales $\subset$ net income
 - *relevance* – important for investors

Tax Accounting

- *direct write-off* (mostly) required because:
 - “*begging game*” – high *AUA* could artificially depress taxable income
 - *reliability* – more important in tax accounting



- 20 -



Allowance Method

▪ Income Statement Approach

- certain percentage of credit sales
- estimated by management based on past experience with collections, customer quality, state of economy, etc.
- drawback is if the company is consistently better (worse) in collecting its receivables than expected *AUA* will be excessive (insufficient)

- 21 -



Allowance Method

▪ Balance Sheet Approach

- desired ending level of *AUA* is first determined and the allowance is incremented so that this level of *AUA* is reached
 - *single percentage* applied to the entire *AR*
 - *aging of accounts* – different percentages for different classes of accounts depending on how much overdue they are, e.g. 1% for <30 days, 3% for 30 – 60 days, 10% for >60 days
 - *specific analysis* of individual accounts

- 22 -



Allowance Method



- **Label**
 - “allowance for uncollectible accounts”, or
 - “allowance for doubtful accounts”, or
 - “allowance for bad debts”
- **Disclosure**
 - on BS receivables are typically disclosed net of allowances for doubtful accounts (see Cisco)
 - detailed disclosure in footnotes

Assets	2001	2000
Accounts receivable, net of allowance for doubtful accounts of 28 in Dec 31, 2001, and 22 in Dec 31, 2000	272	248

- 23 -



Allowance Method



If an invoice is due in 30 days, we pay it in 60 days.
 If it's due in 60 days, we pay in on 90 days. If it's
 due in 90 days, then they probably don't
 need the money anyway.

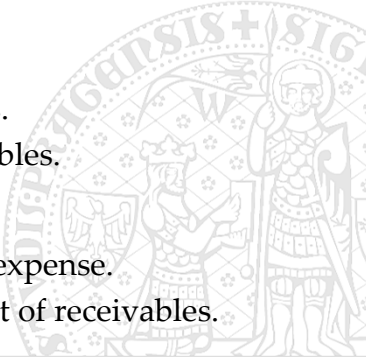
- 24 -



Extra – Allowance Exercise



Allowance Method

- **End of 2000**
 - Firm observes accounts receivable (AR) and sets up allowance of uncollectible accounts (AUA).
 - **During 2001**
 - Firm makes credit sales.
 - Firm writes off receivables.
 - Firm collects some receivables.
 - **End of 2001**
 - Firm recognizes bad debt expense.
 - Firm computes net amount of receivables.
- 



Allowance Method

- **End of 2000**
 - ❶ gross *AR* are 270
 - ❷ out of that *AR* worth 70 are overdue
- **Bad Debt Policy**
 - based on its past experience with uncollectible accounts receivable firm decides to hold
 - 20% *AUA* for overdue *AR*
 - 4% *AUA* for non-overdue *AR*
- **Required**
 - Record gross *AR* and *AUA* for Dec 31, 2000.

- 27 -



Allowance Method

- **Accounts Receivable Gross (2000)**
 - total receivables: $AR_{2000} = 270$
 - overdue receivables: $Overdue_{2000} = 70$
 - non-overdue rec.: $NonOverdue_{2000} = 270 - 70 = \underline{200}$
- **Allowance for Uncollectible Accounts (2000)**
 - $AUA_{2000} = 20\% \cdot Overdue_{2000} + 4\% \cdot NonOverdue_{2000}$
 - $AUA_{2000} = 20\% \cdot 70 + 4\% \cdot 200 = 14 + 8 = \underline{22}$

- 28 -



Allowance Method

▪ **Balance Sheet 2000**

Assets		Liabilities, Equity		Expenses	Revenues
Accounts Receivable	270	Contributed Capital	248		
	AUA				
	22				
	248				

- 29 -



Allowance Method

☞

▪ **During 2001**

- ❶ makes credit sales gross amount of 840
- ❷ writes off AR of bankrupt customer worth 20
- ❸ collects cash for receivables worth 790

▪ **Required**

- Record the transactions of 2001.
- Compute gross AR on Dec 31, 2001.
- Compute temporary level AUA after these transactions.

- 30 -



Allowance Method

Balance Sheet Approach

Assets		Liabilities, Equity	Expenses	Revenues
Accounts Receivable		Contributed Capital		Sales
①	270	①		
①	840	①		840
	②			
	20			
	③			
	790			
AUA				
	④			
	22			
②	20			
①	248			
Cash				
③	790			

- 31 -



Allowance Method

Accounts Receivable – Gross

$$AR_{2001} = AR_{2000} + CreditSales_{2001} - WriteOffs_{2001} - CashCollections_{2001}$$

$$AR_{2001} = 270 + 840 - 20 - 790 = \underline{300}$$

Allowance for Uncollect. Accounts – Temporary

$$AUA_{Temp} = AUA_{2000} - WriteOffs_{2001}$$

$$AUA_{Temp} = 22 - 20 = \underline{2}$$

- 32 -



Allowance Method

- **End of 2001**
 - makes **no changes** to its bad debt expense policy
 - observes that **one third** of AR is overdue
 - ④ recognize bad debt expense for year 2001
- **Required**
 - Compute desired level of AUA on Dec 31, 2001.
 - Compute the bad debt expense for 2001.
 - Record the bad debt expense for 2001.
 - Compute net AR on Dec 31, 2001.
 - Report AR on the balance sheet of Dec 31, 2001.

- 33 -



Allowance Method

- **Required Allowance 2001**
 - $AUA_{2001} = 20\% \cdot Overdue_{2001} + 4\% \cdot NonOverdue_{2001}$
 - $AUA_{2001} = 20\% \cdot 100 + 4\% \cdot 200 = 20 + 8 = \underline{28}$
- **Bad Debt Expense 2001**
 - $Ex_{2001} = AUA_{2001} - AUA_{Temp}$
 - $Ex_{2001} = 28 - 2 = \underline{26}$
- **Accounts Receivable – Net**
 - $netAR_{2001} = AR_{2001} - AUA_{2001}$
 - $netAR_{2001} = 300 - 28 = \underline{272}$

- 34 -



Allowance Method

▪ **Balance Sheet Approach**

Assets		Liabilities, Equity		Expenses	Revenues
Accounts Receivable		Contributed Capital			Sales
0 270		0 248			1 840
1 840					
	2 20				
	3 790				
AUA					
	0 22				
2 20				Bad Debt Expense	
	4 26			4 26	
4 272					
Cash					
3 790					

- 35 -



Allowance Method

▪ **Label**

- “allowance for uncollectible accounts”, or
- “allowance for doubtful accounts”, or
- “allowance for bad debts”

▪ **Disclosure**

- on BS receivables are typically disclosed net of allowances for doubtful accounts (see Cisco)
- detailed disclosure in footnotes




Assets	2001	2000
Accounts receivable, net of allowance for doubtful accounts of 28 in Dec 31, 2001, and 22 in Dec 31, 2000	272	248

- 36 -

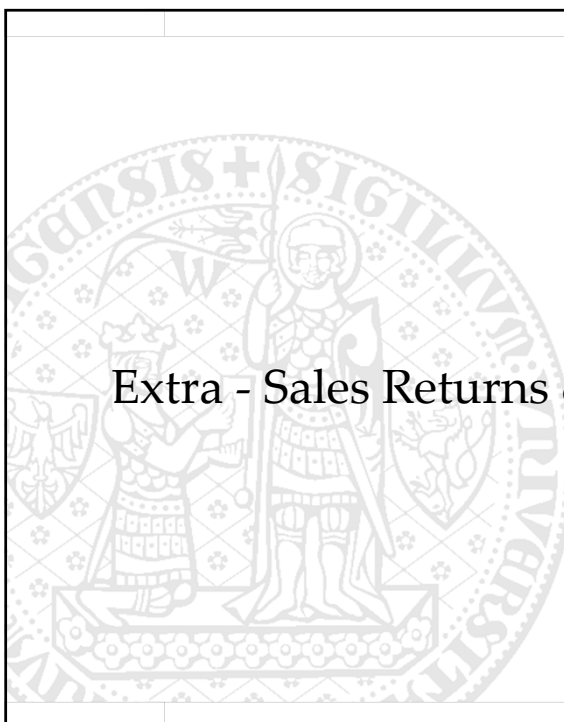


Allowance Method



If an invoice is due in 30 days, we pay it in 60 days.
If it's due in 60 days, we pay in on 90 days. If it's
due in 90 days, then they probably don't
need the money anyway.

- 37 -



Extra - Sales Returns & Allowances



Sales Returns

- **Occurrence**
 - customers can cancel their purchase or request a price concession in case the delivered goods are damaged or do not meet the pre-contracted parameters
 - this is essentially a reduction in previously contacted sales and that is also how it is treated

- 39 -



Sales Returns

- **Treatment**
 - “sales returns and allowances” (*SRA*) is a contra-revenue account offsetting sales (that is why we refer to “net sales”)
 - the treatment to bad debts, but for the fact that the returned goods increase inventory and thereby decrease *COGS*
 - *AR* or cash are credited (reduction in cash or receivables), *SRA* is debited (reduction in sales)
 - *Iv* are debited (goods returned), *COGS* are credited (reduction in matching costs)

- 40 -



Sales Returns

Transaction Analysis

- ① merchandise purchased for \$ 720 sold to a customer for \$ 800 (no sales returns expected)
- ② one quarter of the goods is returned because it has been shipped to a wrong customer

Assets		Liabilities, Equity	Expenses		Revenues		
Accounts Receivable			COGS		Sales		
❶	800		❶	720		❶	800
	❷		200		❷	180	
Inventories					SRA		
	❶	720			❷	200	
❷	180						

- 41 -



Sales Returns

Allowance for Sales Returns and Allowances

- ① at the time of sale estimated *SRA* is recorded and an contra-asset allowance for sales returns and allowances (*ASRA*) is created
- ② when goods are returned *AR* are eliminated against *ASRA* (i.e. no IS effect)

Assets		Liabilities, Equity	Expenses	Revenues	
	Accounts Receivable			Sales	
①	800				① 800
②					
	ASRA			SRA	
①		40		① 40	
②	30				
② ①	760				

- 42 -



Accounts Receivable





"You handled the audit so well, we're promoting you from Accounts Receivable to Accounts Deceivable."

- 43 -